Introduction to Environmental Engineering

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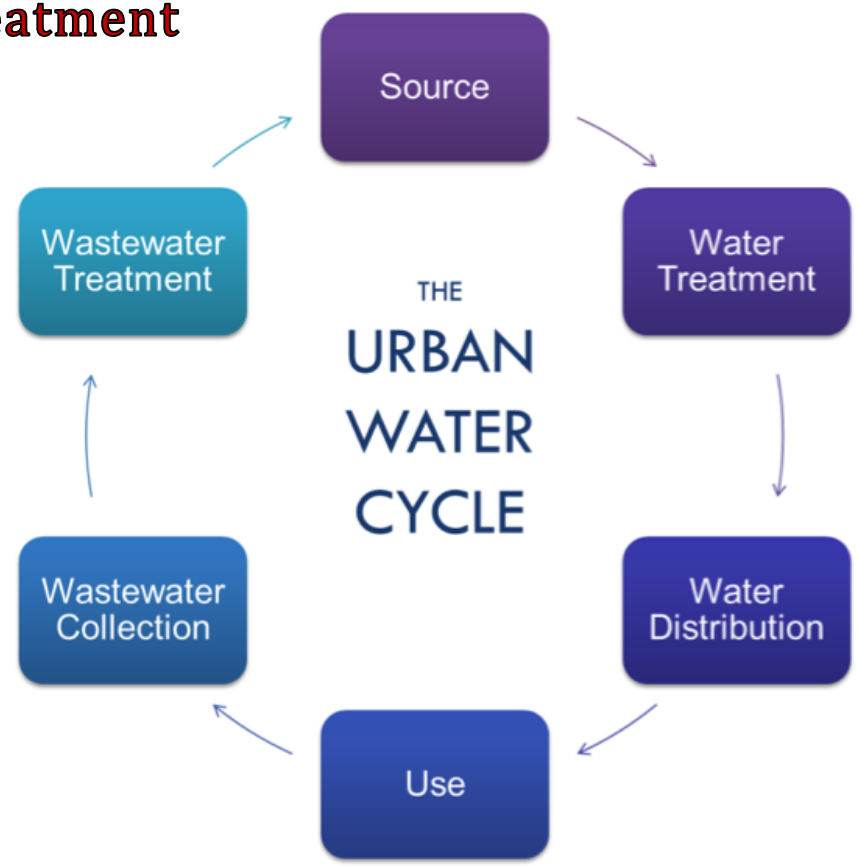


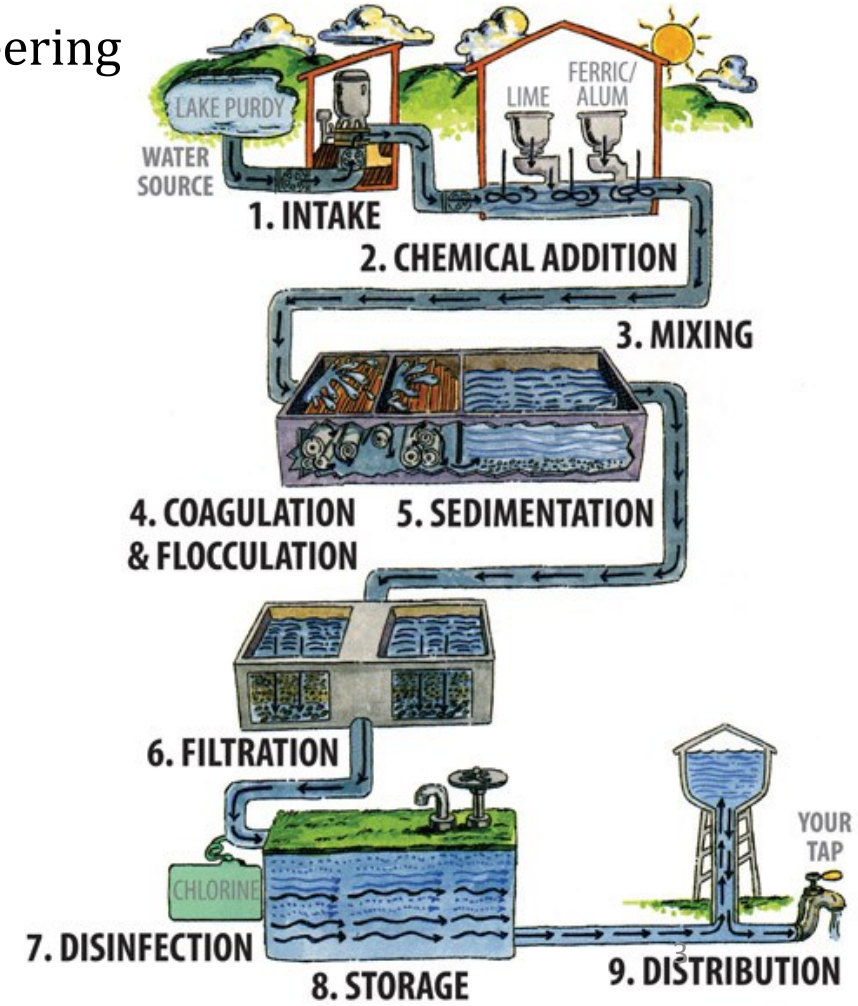

Construction Management
Structural Engineering
Earthquake Engineering
Transportation Engineeing
Civil Engineering
Costal Engineering
Water Resource Management
Environmental Engineering
Geotechnical Engineering

Introduction to Environmental Engineering

- There are several divisions in the field of environmental engineering

1. **Water Supply & Treatment**

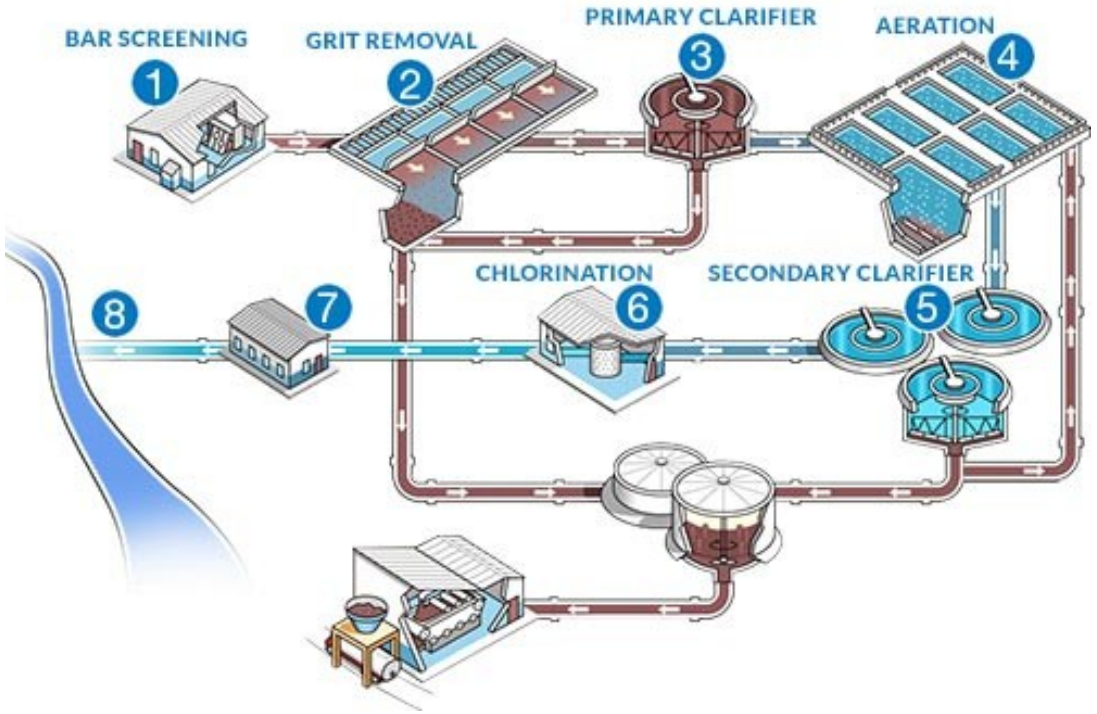




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2. Wastewater Collection & Treatment

Environmental Engineers develop collection & treatment systems to discharge wastewater safely into the environment



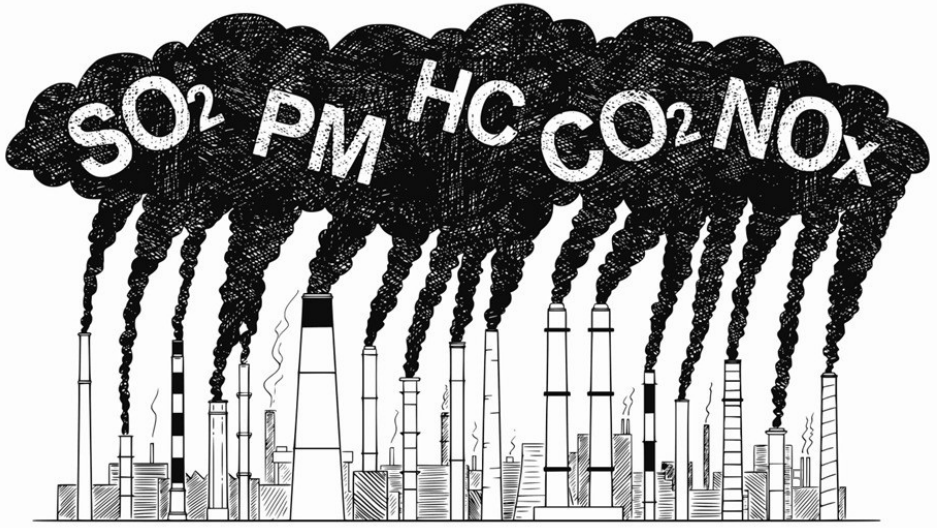
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3. Air Quality Management

Develop/employ technologies to reduce/eliminate the emission of harmful substances into the atmosphere

Develop dispersion models to evaluate the concentration of a pollutant at a receptor source or the impact on overall air quality

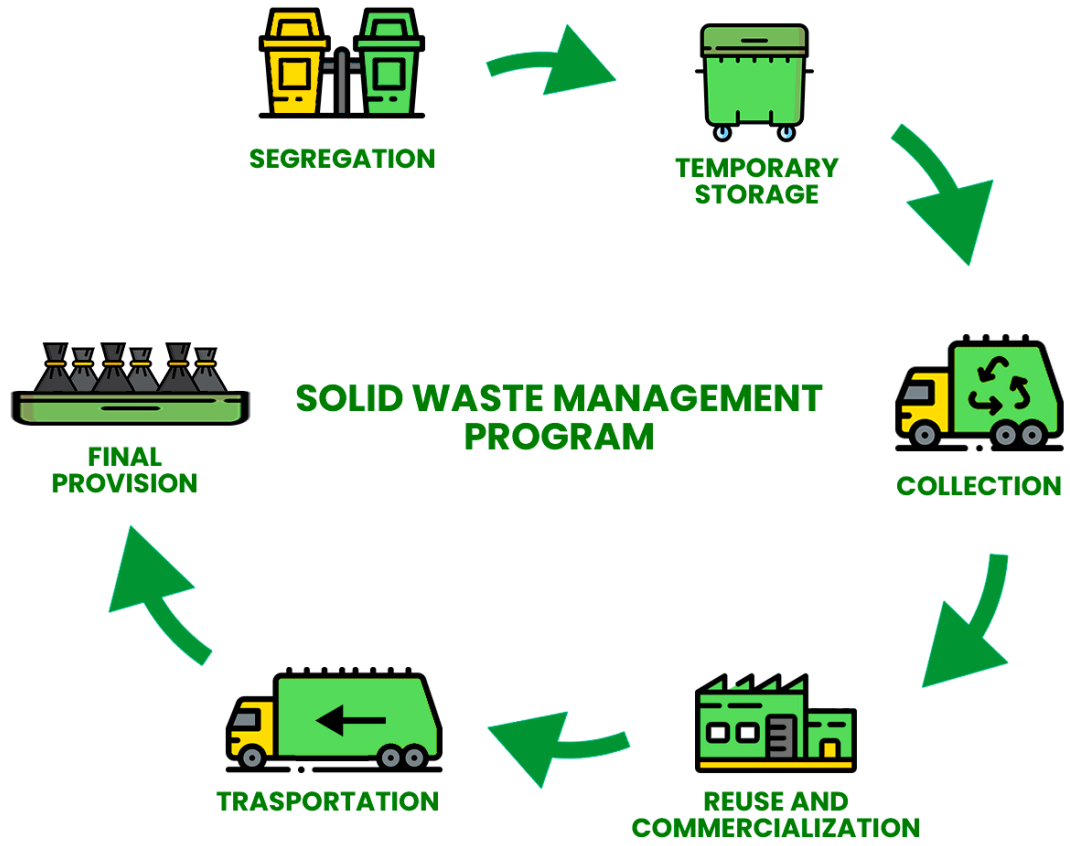
Environmental engineers also attempt to prevent the production and repercussion of acid rain, global warming, ozone depletion etc.



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4. Solid Waste Management

Controlling the generation, storage, collection, transport or transfer, processing and disposal of solid waste materials



5. Hazardous Waste Management



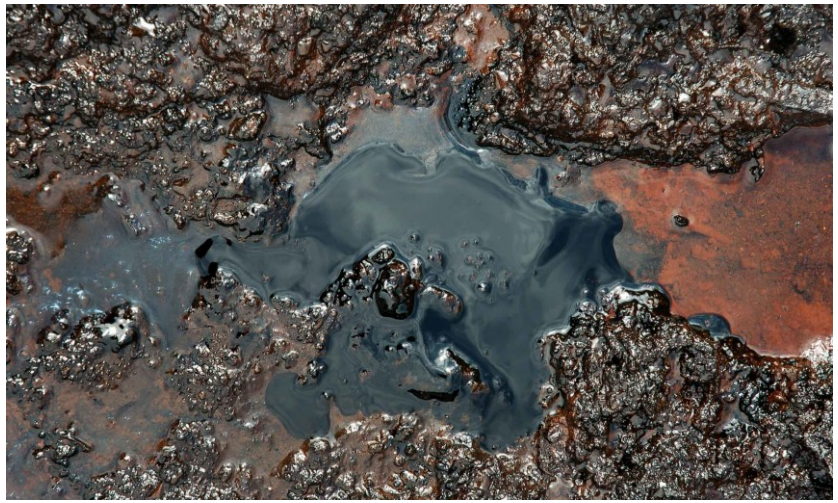
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6. Industrial Waste Management

Design new processes or modify existing processes to reduce or eliminate the release of pollutants

7. Contaminated Land Management & Site Remediation





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8. Environmental Health and Sanitation





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9. Environmental Impact Assessment and Mitigation

Environmental engineers assess the impacts of a proposed project on the environmental

- **Water quality - Air quality - Habitat - Flora & fauna - Agricultural capacity**
- **Traffic impacts - Social impacts - Ecological impacts - Noise impacts - Visual impacts**

If impacts are expected, they then develop mitigation measures to limit or prevent such impacts

10. Environmental Policy and Regulation Development

Introduction to Environmental Engineering

- Environmental engineering is a synthesis of various disciplines
- It incorporates elements from;



Civil Engineering



Law



Public Health



Mechanical Engineering



Chemistry



Biology



Geology

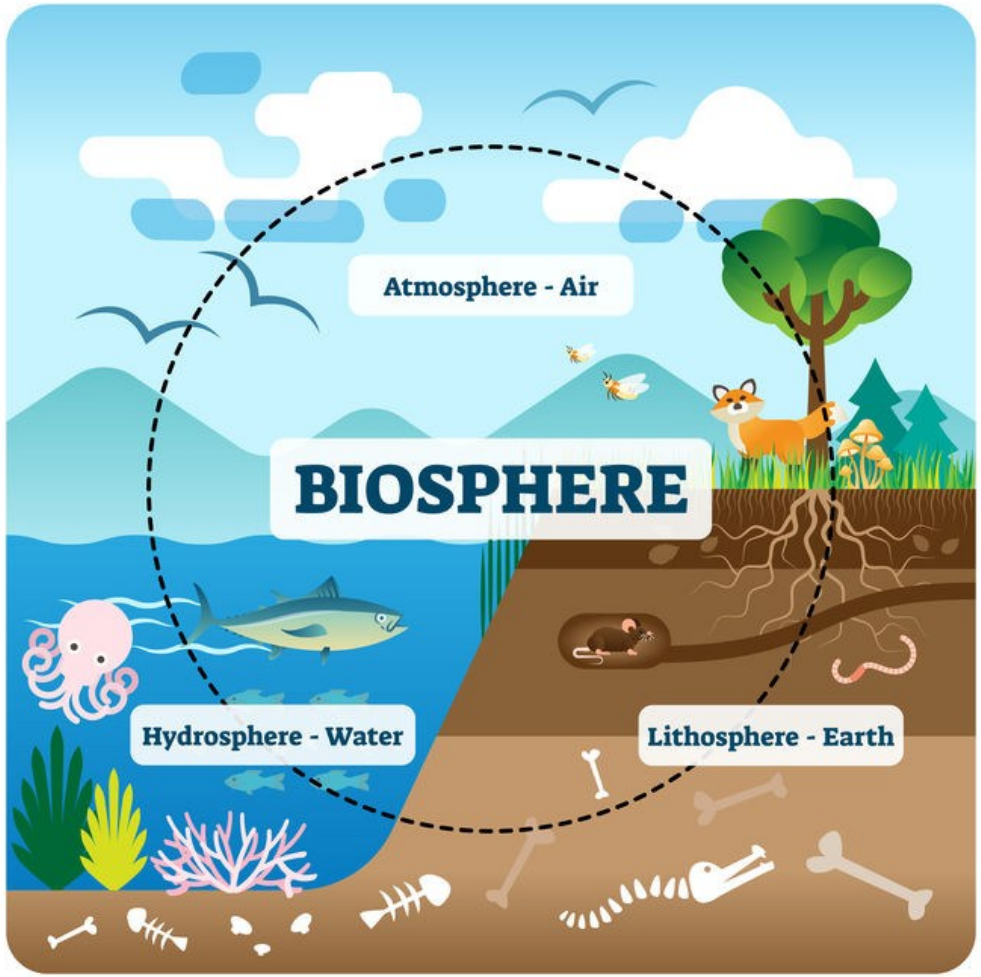


Ecology

Environment and its Components

- Our Earth is made up of 4 spheres

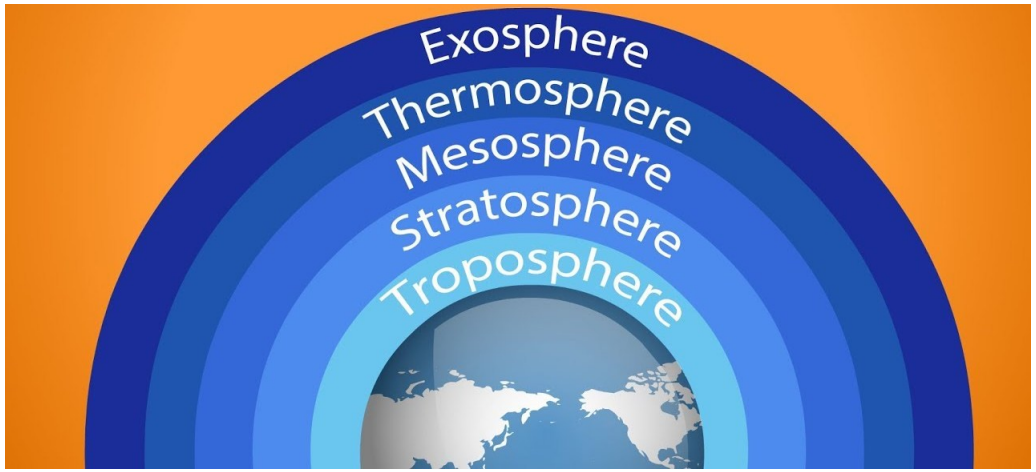
- Atmosphere
- Lithosphere
- Hydrosphere
- Biosphere



Environment and its Components

Atmosphere

- This is the gaseous envelope surrounding the Earth
- It extends up to 1000 km above the Earth surface
- The atmosphere protects the Earth from dangerous UV radiation, provides life sustaining Oxygen, CO_2 needed for photosynthesis and macronutrient Nitrogen
- The atmosphere plays a major role in maintaining the heat balance of the Earth
- The atmosphere is the medium of carriage of water from oceans to land in the hydrological cycle



Environment and its Components

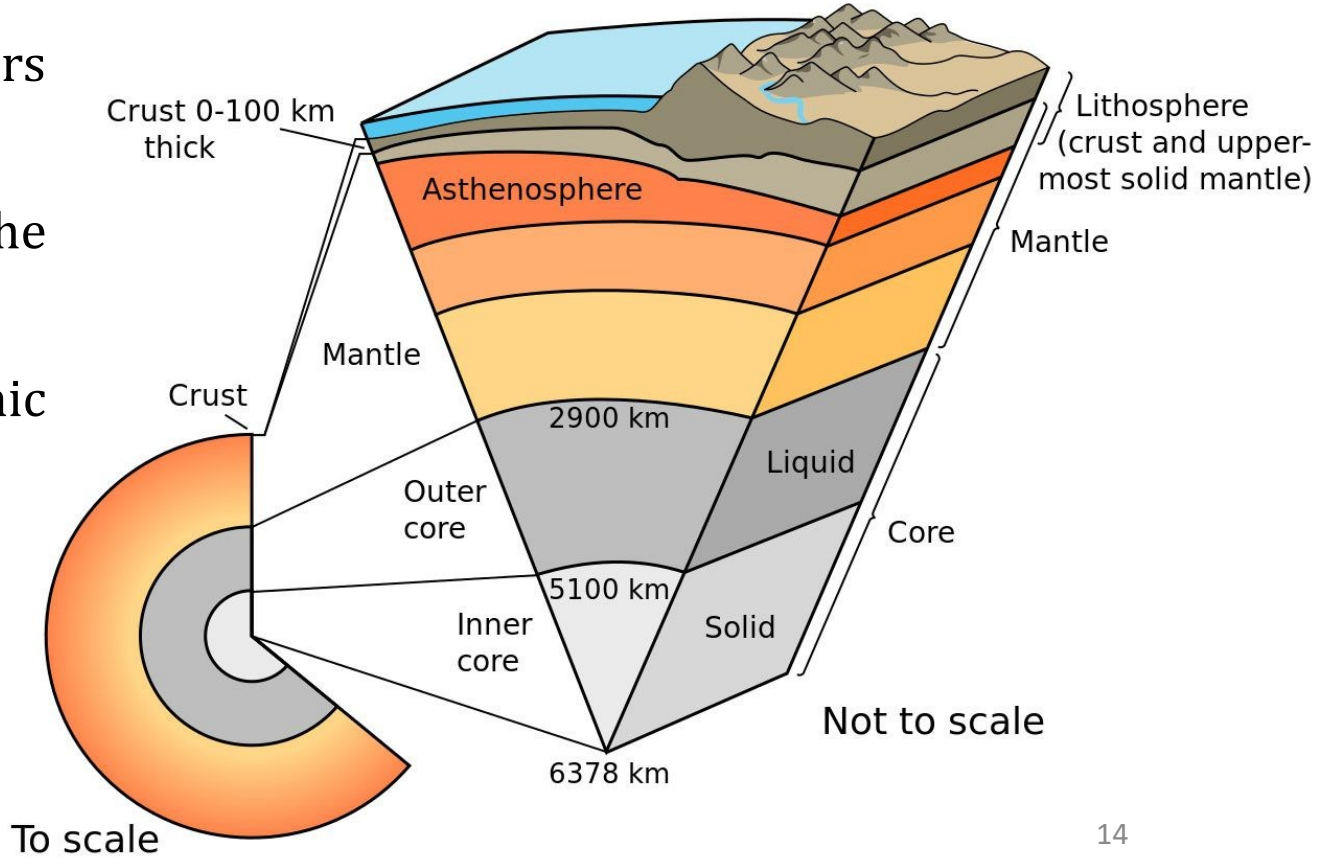
- The composition of the atmosphere is as follows (2017)

Constituent	Volume (%)
Nitrogen	78.1%
Oxygen	20.9%
Water Vapor	0.1 – 5%
Argon	0.9%
CO_2	0.04%
Trace Constituents	balance

Environment and its Components

Lithosphere

- The lithosphere is the solid, outermost layers of the Earth's structure
- It includes the brittle upper portion of the mantle and the crust
- The soil is made up of organic and inorganic matter and water



Environment and its Components

Hydrosphere

- The hydrosphere consists of all water resources on Earth, both surface water and groundwater
- The world's water is found in ocean's and seas, lakes and reservoirs, rivers and streams, glaciers and snowcaps, in groundwater and in the air.
- Water moves through the hydrosphere in a cycle

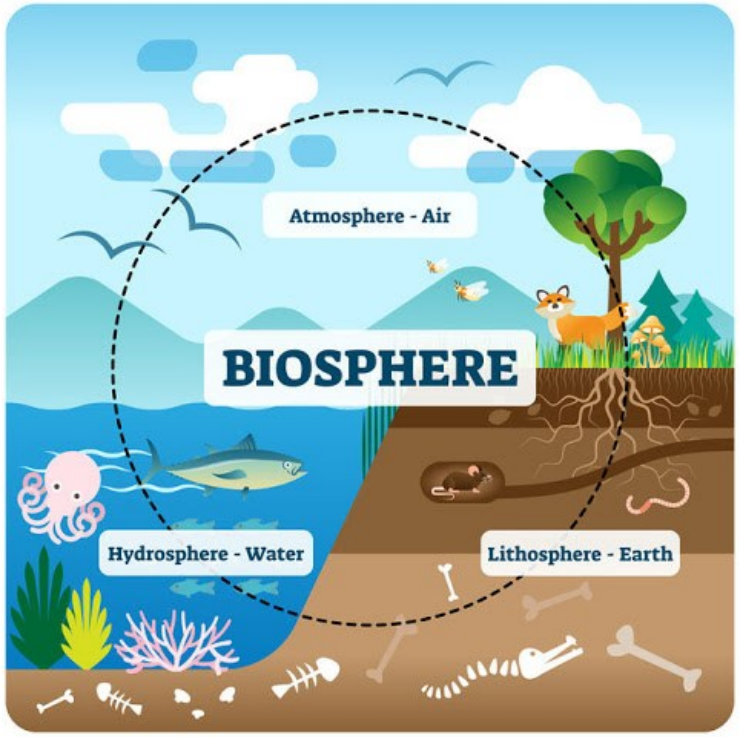


Oceans and Seas	**96 - 97%**
Glaciers and Polar ice caps	**2 - 3%**
Fresh water	**<1 %**

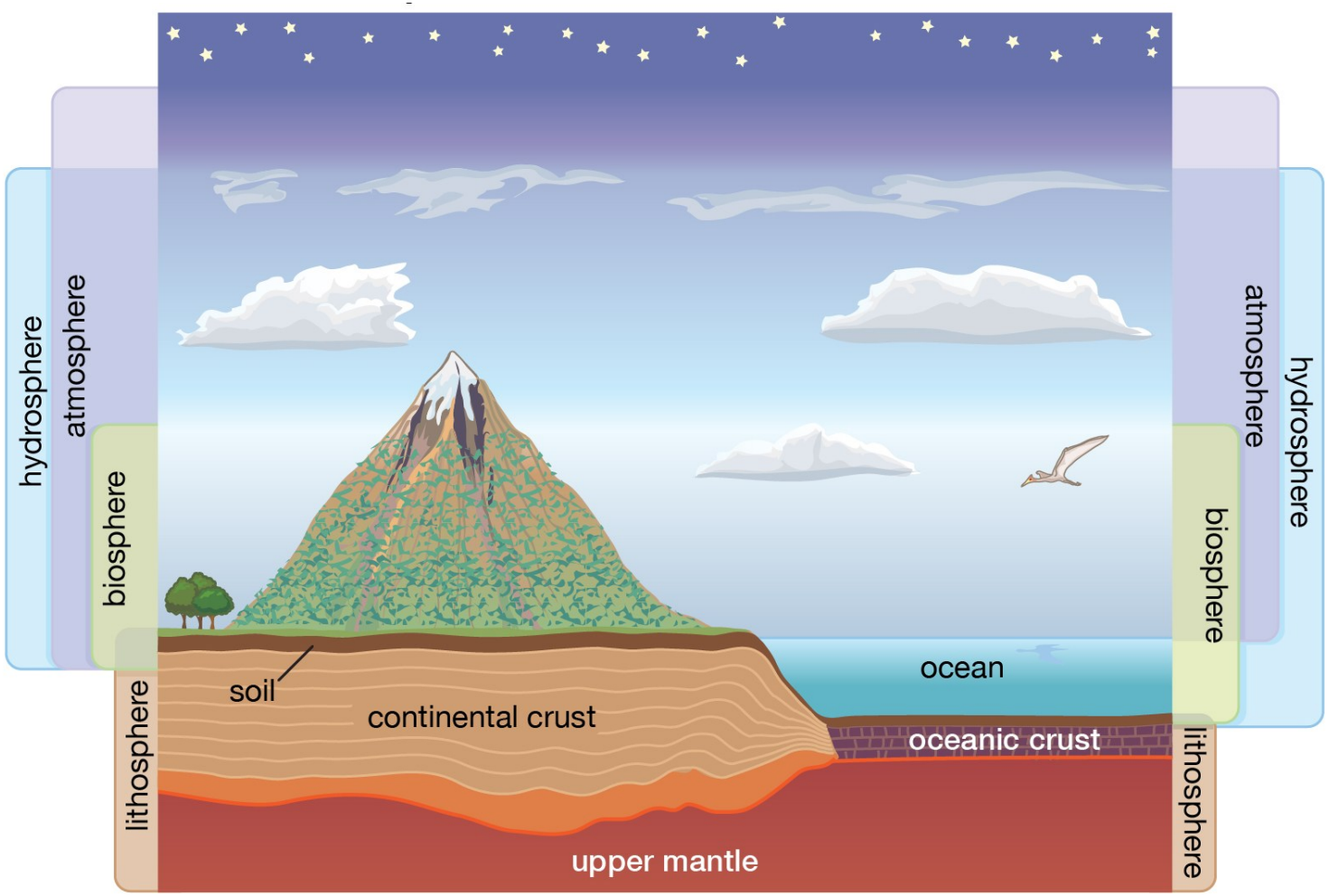
Environment and its Components

Biosphere

- Biosphere is the capsule encircling the Earth's surface wherein all living things exist
- The biosphere extends 10 km below sea level to 10 km above sea level
- Life forms do not exist outside this zone
- The biosphere covers parts of other segments of the environment (Lithosphere, Hydrosphere & Atmosphere)



Environment and its Components

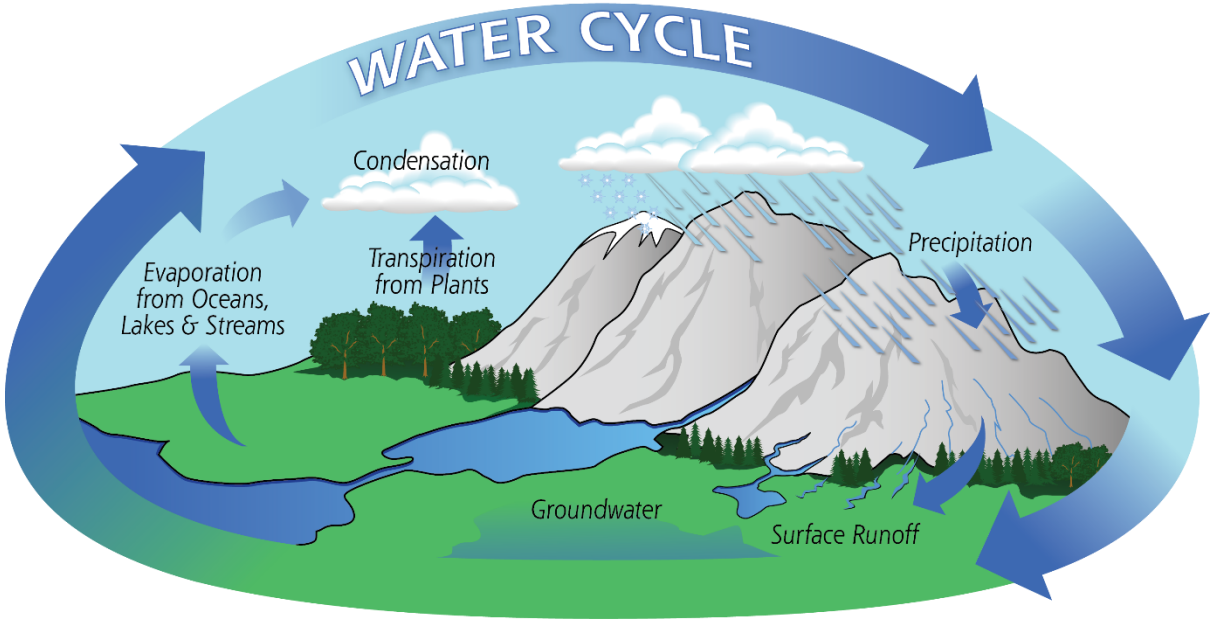


Material Cycles

- There is a cyclic movement of nutrient materials between the biotic and abiotic environment
- The most important and common cycles are

Water Cycle

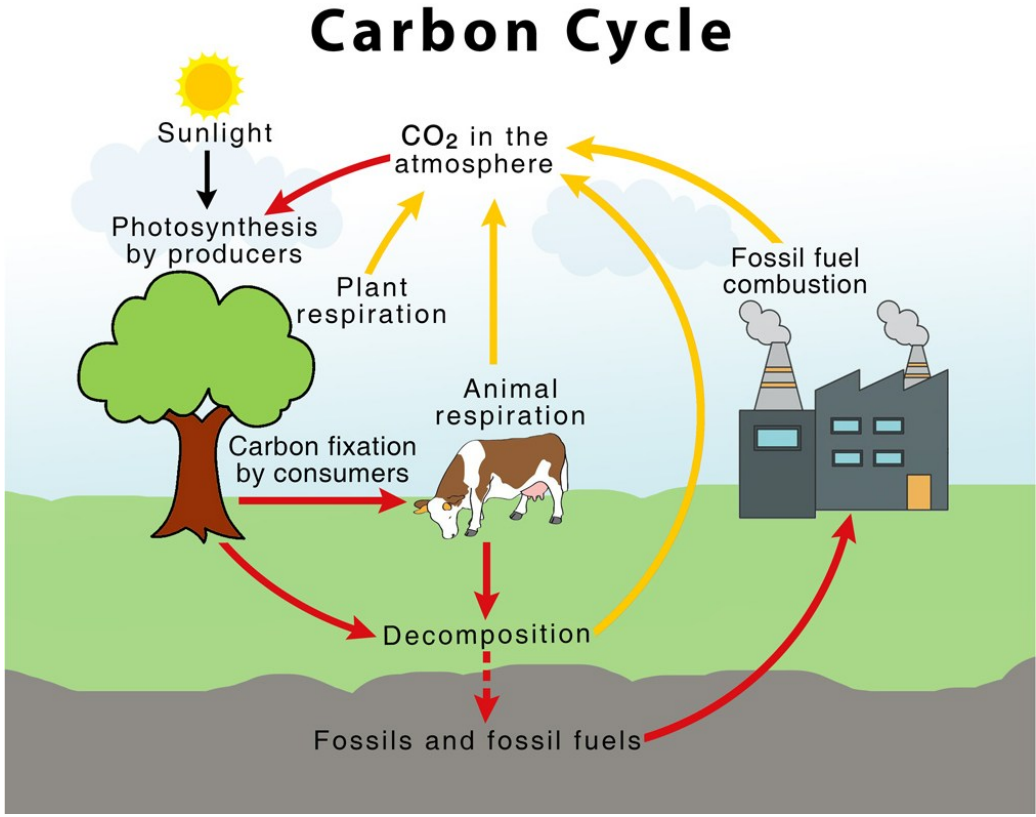
- There is a constant exchange of water between air, land, sea and living beings
- The water cycle is driven by solar energy
- This is the main source of fresh water for the living beings



Material Cycles

Carbon Cycle

- Carbon is an essential component of all plants, animals and organic matter
- Human activities have a massive impact on the carbon cycle
- Burning fossil fuels and changing land use transfer significant quantities of carbon into the atmosphere
- The CO_2 concentration in the atmosphere is greater than anytime in the last 3.6 million years

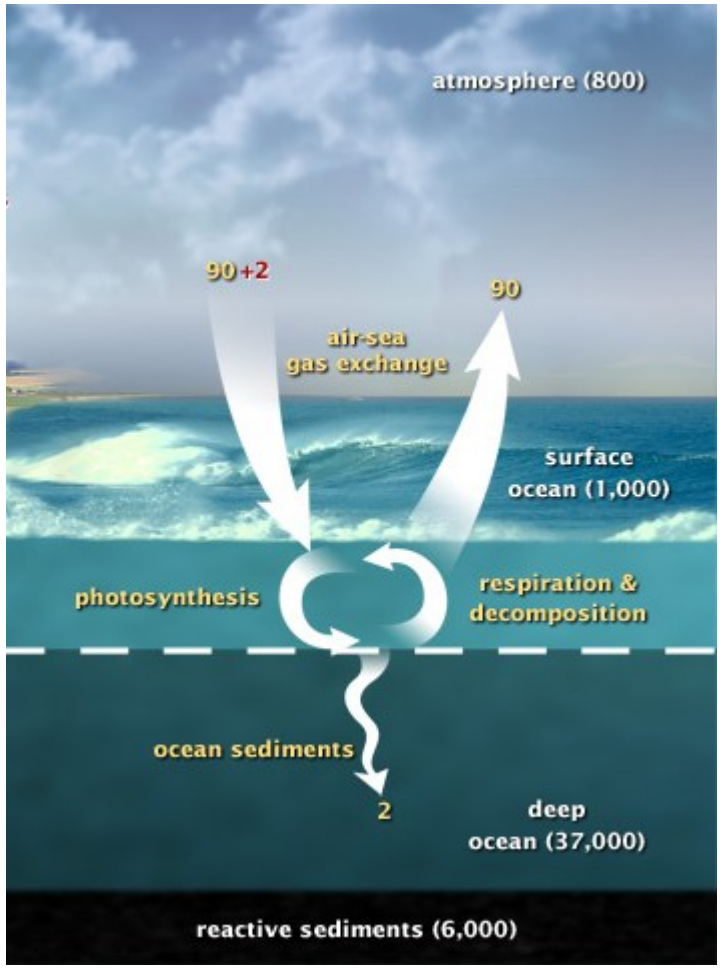


Material Cycles

Carbon Cycle

- The ocean absorbs much of the CO_2 in the atmosphere and thus play a vital role in regulating the carbon cycle
- This extra CO_2 is lowering the ocean's pH, and acidifying our oceans





Material Cycles

Nitrogen Cycle

- Nitrogen is available in the atmosphere cannot be directly absorbed by plants
- In order to be absorbed by plants it has to be converted to water soluble compounds
- This is called fixation of nitrogen
- Plants can absorb nitrogen in the form of NH_4^+, NO_3^-

